

Exam 1: Basics of R – Version D

Exam Agreement

This exam is an **individual assessment** and you should neither offer help to nor receive help from another person.

Cell phones are to be handed in to the instructor and all other electronic devices must be stored in bags at the front of the room.

No smart glasses or other wearable electronic devices are to be powered on during this exam except with accommodations.

You may use **no physical or electronic** resource to complete the work.

By signing, you agree that you have neither given nor received unauthorized aid on this examination.

Printed Name: _____

Chapman ID: _____

Signed: _____

Section: _____ **Date:** _____

Include this signed page as the first page of your submitted work.

Exam Instructions

You must answer every question on the exam to the best of your ability. Include all appropriate code, syntax, and functions that would lead to the code to run successfully.

Question 1: How Computers Work

What are four things that all computers have in common? Describe each one briefly.

Question 2: Command line and syntax

Describe how and why square brackets are used in the R language. Give an example of code that uses these brackets.

Question 3: Data Frames

In the data frame `beets` (Fig. 1), what is the standard deviation of sugar percentage in harvest 2?

beets	30 obs. of 5 variables
\$ harvest:	Factor w/ 2 levels "harv1","harv2": 1 1 :
\$ block :	Factor w/ 3 levels "block1","block2",...:
\$ sow :	Factor w/ 5 levels "sow1","sow2",...: 3 4 :
\$ yield :	num 128 118 95 131 136 ...
\$ sugpct :	num 17.1 16.9 16.6 17 17 17 17 16.7 16.4

Figure 1: The beets data set.

Question 4: Matrices and Arrays

Create a matrix containing random numbers. Each dimension should be greater than 2 in size.

Question 5: Lists

Using the `wasp.trees` data set (Fig. 2), how would you reference the integer 14?

 wasp.trees	List of 2	
List of 2		
\$:list of 4		
..\$ edge	: int [1:36, 1:2]	20 20 21 22 22 23 24 24 2...
..\$ Nnode	: int	18
..\$ node.label:	chr [1:18]	" " " " " " " " ...
..\$ tip.label :	chr [1:19]	"P._regalis" "P._spec._nov._1"...
..- attr(*, "class")=	chr	"phylo"
..- attr(*, "order")=	chr	"cladewise"
\$:list of 4		
..\$ edge	: int [1:28, 1:2]	16 17 18 19 19 18 20 20 2...
..\$ Nnode	: int	14
..\$ node.label: chr	[1:14]	" " " " " " " " "77" ...
..\$ tip.label :	chr [1:15]	"S._4_{obliqua}" "S._2_{aff._o..."
..- attr(*, "class")=	chr	"phylo"
..- attr(*, "order")=	chr	"cladewise"

Figure 2: The `wasp.trees` data set.

Question 6: Class Coercion

If you run the following lines:

```
a <- c(0, 1, 0, 0)
all(a)
```

You will get a warning, but `all()` will return `FALSE`. Explain why this works, even though `all()` can only take logical values.

Question 7: Data Classes and Special Values

You are trying to reference the sugar percentage in the data set `beets` (Fig. 1) using the following line:

```
beets$sugrpct
```

R returns `NULL`. This is clearly not right. Why does it return this and how can you fix it?

Question 8: Factors

Make a vector of ordered factors describing how fearful you are of the following things: clowns, spiders, snakes, cockroaches, dungeons. Your ordered factor must have at least three levels. Store this vector as `scary_things`.

Question 9: Vectors

The following is not a valid vector in R. Describe why this is the case.

```
vec <- c("yes", 8, 9, 1.1, NA)
```

Question 10: Troubleshooting

You are running a friend's code and get the following error:

```
Error: object 'my_list' not found
```

What can you tell your friend to help them fix this error?